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1

Introduction

In the 21st century, the global economy seems to be facing countless global crises, from ecological overshoots to worsening climate change, extreme human deprivations, and increasing inequality. These challenges are as closely related to society's trend of consumption and production of goods and services as political economy and its components. Ever since the 1950s, economists have observed a rather compelling result in growth theory, termed a natural resource curse (Brunnschweiler et al., 2008). The natural resource curse is a negative relation between natural resource abundance and growth where resource-rich countries grow slower than resource-scarce countries. Auty et al., 1993 in the context of oil-exporting countries to describe this puzzling phenomenon coined it. The crucial question, for instance, is why countries like Norway and Botswana can do well, while Venezuela and Angola cannot. While it can depend on the nature of the non-renewable resource, one major reason seems to be concerned with how these natural resources are managed. The economic theory highlights the critical role of institutions in determining how natural resources are managed. Institutions shape property rights, establish regulatory frameworks, enforce contracts, promote transparency and accountability, influence institutional quality, and engage local communities, (Acemoglu et al., 2005) seem to influence natural resource management to a large extent.

In this paper, we re-examine the curse literature and its impact after including institutional quality by two mediums. The first medium is macro-modeling through which we look at two states- 'curse' and 'blessing'. Curse is understood as the negative relation between growth and natural resource use, and blessing is when the interaction has a positive relation (by including institutions). We analyze the central question by augmenting the Solow green model by incorporating institutional quality. The central question is, "How does the quality of political institutions influence the impact of natural resource abundance on economic development in resource-abundant countries?" The second medium is to empirically examine the results of the literature. Most studies investigating the institutional association to curse find similar results through the threshold regression analysis. In closing, we revisit Sarmidi et al., 2014 threshold regression analysis and Gylfason, 2001 education component. In the modeling section, we begin by deriving the predictions of the Solow model on long-term balanced growth, to further look more closely at Romer's extension of the Solow model with environmental aspects, where he included natural resources. We further extend the Green Solow Model by adding the effectiveness of natural resources, which acts as a proxy for institutional quality, and argue that de-growth prospects posed by the natural resource can be outweighed by this parameter. It is not within the scope of this paper to realize the components of institutional quality in the modeled parameter. The aim is to reassess the results through an uncharted framework. In the last section, we discuss some implications of solow assumptions and probable shortcomings of the model.

In the literature section, we take a closer look at natural resource abundance and natural resource curse studied and understand the importance of considering institutional quality in various frameworks.

2

Literature Review

The varied literature on the "resource curse" has understood the significance of natural resources and their economic potential quite diversely. Although the literature is quite immense and has contributed to growth and development studies, recent works have shown a more nuanced and not straightforward correlation. The literature has no single commonly accepted definition, but different definitions share the basic concept of the reduced growth rate of resource-rich countries compared to resource-poor. The literature essentially studies two major directions. First, the causality between growth and natural resource abundance, and relation with institutional quality. The reverse causality between economic growth and natural resource abundance has been addressed by J. Sachs et al., 1995; J. D. Sachs et al., 1999. Arguably so, rich countries with great natural resource wealth might dodge the trend, nevertheless, the data provides insight into slower growth in resource-rich to– poor countries.

In the literature, the negative trend phenomena have been understood by numerous channels. Corden et al., 1982 and Auty et al., 1993 realized Dutch disease, the phenomenon wherein the surplus wealth generated from the sale of natural resources led to an appreciation of the real exchange rate, subsequently causing a contraction in the traded sector. Gylfason, 2001 identified breeding corruption in business and government which reduces economic and social efficiency and neglects

education and human development as channels leading to de-growth prospects. Similarly, while the political intervention in resource endowment can generate an understanding of whether the natural resource is a curse (Robinson et al., 2006), the emphasis must be on rapid economic development through high human resource expenditure, and fostering education(Levine, 1998). Further, Knack et al., 1995 provides insights on institutional quality predominant in achieving long-term growth through investment all of which hinted that factors of Dutch disease, rent-seeking, and investments in education can be intervened by the institutions and their role of governance.

Cabrales et al., 2011 in his paper *The Quality of Political Institutions and the Curse of Natural Resources* puts forward a theoretical model that emphasizes the behavior and incentives of politicians, through a standard voting model with the possibility of revolution, to explain regularities related to the curse of natural resources. Further, it is corroborated with empirical findings. Their results suggest that in the presence of poor political institutions, human capital tends to diminish concerning natural resources, whereas in environments characterized by high institutional quality, this relationship is reversed. Qiang et al., 2020 study of cross-country sample of China analyses the province-level sample and focuses on resource abundance, institutional quality, and economic growth. Using the varying coefficient method, the authors find that resource abundance has a positive effect on economic growth at a provincial level in China between 1990 and 2008 and this effect depends non-linearly on institutional quality.

Brunnschweiler et al., 2008 re-examines the effects of natural resource abundance on economic growth using different measures of resource endowment and finds a positive direct empirical relationship between natural resource abundance and economic growth without controlling for institutional quality insinuating the non-linearity of relation. The result by Mehlum et al., 2006 also rejected the curse hypothesis partially. The author in an attempt to study the institutional influence proposes that high-quality institutions can be categorized

into two types: grabber-friendly and producer-friendly. In their model, grabber-friendly institutions influence the resource being cursed while producer-friendly has a less pronounced effect. Further, the grabber-friendly institutions lay the foundation for rent-seeking activities and other economically inefficient tendencies like capturing and exploiting natural resource wealth for their benefit, corruption, and economic distortions. In environments with producer-friendly institutions, it is anticipated that resource-rich nations will draw in more producers to engage in economic growth, fostering greater investment, innovation, and overall economic expansion. Conversely, in settings with grabber-friendly institutions, the centralization of resource wealth among a select elite obstructs wider economic progress and perpetuates the phenomenon known as the resource curse. In summary, Mehlum et al., 2006 finds that the impact of natural resource abundance on economic development depends on the quality of institutions.

In closing, although the literature has thoroughly investigated the role of institutions in analyzing the natural resources to growth trends, nonetheless, the effect of institutional quality in the Solow growth model with natural resources remains unexplored.

3

Model

In this section, we introduce the problem of resource limitation through the Solow growth framework and look at the growth parameter. This exercise follows from the Solow, 1956. We start with the Solow model because it is a fundamental model which aids in simplifying assumptions. The Solow model takes into account two factors, namely capital and effective labour and investigates the effects of the division of output between consumption and investment. Production function in the basic Solow model takes the form:

$$Y(t) = K(t)^{\alpha}(A(t)L(t))^{1-\alpha}$$

where output Y depends on capital (K), Labour (L) and technological progress (A). Assumption of competitive markets hold. Households own the labour and supply in-elastically. The model also holds the assumption of constant returns to scale. In this model government is absent and exogenous factors such as rates of saving, depreciation, population growth, and technological progress are assumed constant. Since it is a continuous time model, variables are defined in time and parameters grow at a constant rate which is given. For a variable X, the growth rate is,

$$\frac{\dot{X}}{X} = g_X$$

Similarly, we state the dynamics of the variables Labour (L) and Technology (A) following Romer, 2012

$$\frac{\dot{L}}{L} = n \qquad \frac{\dot{A}}{A} = g \tag{3.1}$$

The dot denotes the derivative with respect to time. Both n and g are exogenous parameters. And, output is divided between consumption and Investment. Capital (K) depreciates at rate δ. Thus we have motion for capital equation,

$$\dot{K} = sY(t) - \delta K(t) \tag{3.2}$$

Further, in our framework, we want to incorporate natural resources. Thus, we look at the augmented Cobb-Douglas production function of the Green Solow Model (GSM) to include natural resource (R), which is a non-renewable resource used in the production of resources. Further, we denote E(t) as the effectiveness of resources' use. E(t) is a proxy for the government's effectiveness in resource allocation and resource use. In the GSM, the amount of land (T)is also included, which we eliminate for simplicity. Note that this exclusion does not have any economic implications for the problem we are looking at.
The new production function takes the following form:

$$\begin{aligned} Y(t) = K(t)^{\alpha}(E(t)R(t))^{\beta}(A(t)L(t))^{1-\alpha-\beta} \\ \alpha > 0, \beta > 0, \alpha + \beta < 1 \end{aligned} \tag{3.3}$$

The dynamics for capital and labour remain unchanged, but for resources and effectiveness of natural resources, we need to state the dynamics. Resources used in production, denoted as (R), diminish at a negative rate over time due to their fixed quantity, primarily consisting of non-renewable resources. The growth rate of natural resources takes the form:

$$\frac{\dot{R}}{R} = -b \qquad b > 0 \tag{3.4}$$

The effectiveness of resource utilization at time t, represented by E(t) experiences growth commensurate with the rate of effectiveness of the government. This is because government efficiency serves as a crucial

metric for evaluating the natural resource utilization towards growth. We assume that E(t) grows at an exogenous constant rate.

$$\frac{\dot{E}}{E} = p \tag{3.5}$$

Along a balanced growth path in the simple Solow model, that is without E and R, the growth rate depends on the rate of labour and technology, that is n+g. This is the result from Romer, 2012 model. Since all the parameters are strictly positive, the growth rate is always positive. But such is not the case in (GSM). The BGP of the GSM takes the form:

$$g_{Yg}^{BGP} = \frac{(1-\alpha-\beta-\gamma)(n+g)-\beta b}{1-\alpha}, 1-\alpha > 0 \tag{3.6}$$

The growth rate in this case can be positive or negative as we have a de-growth prospect by parameter β. This can be identified as a resource curse, where the incorporation of natural resources leads to de-growth. The constraints imposed by limited non-renewable resources may result in a decline in output per worker over time. However, it is important to note that this decline is not inevitable and can be influenced by various factors, one of which we are interested in is the government's institutional quality. (Another can be a technological shock). Without including the government in the model, we are only interested in the effect of good institutional quality.

Next, we solve our model with effective use of resources. We first take the log of equation (3.3),

$$\ln Y(t) = \alpha \ln K(t) + \beta(\ln E(t) + \ln R(t)) + (1-\alpha-\beta)(\ln A(t) + \ln L(t)) \tag{3.7}$$

We can now differentiate both sides of this expression with respect to time. Using the fact that the time derivative of the log of a variable equals the variable's growth rate, we obtain,

$$g_Y(t) = \alpha g_K + \beta(g_R + g_E) + (1-\alpha-\beta)(g_A + g_L) \tag{3.8}$$

Substituting exogenous values of growth rates in the equation from (3.1,.4,.5) in (3.8), we get,

$$g_Y = \alpha g_K + \beta(-b + p) + (1 - \alpha - \beta)(n + g) \tag{3.9}$$

We can now use the fact that g_Y and g_K must be equal if the economy is on a balanced growth path. Imposing $gK = gY$ and solving for g_Y gives us:

$$g_{Y^*}^{BGP} = \frac{(1 - \alpha - \beta)(n + g) - \beta(b - p)}{1 - \alpha}, 1 - \alpha > 0 \tag{3.10}$$

where $g_{y^*}^{BGP}$ is the growth rate of Y on a balanced growth path. We can see that eliminating natural resource R and effectiveness parameter E will bring us back to the result of the Solow model in the basic case. If resources and land were excluded from the model, the growth rate of output Y would align precisely with the growth rate of capital stock K, which is n+g, thereby maintaining the capital stock constant. Further, excluding parameter E, we get a similar result as (3.6) of GSM. We have already discussed what happens when negative b is too high. Thus (3.10) further brings effective use of resources which can reduce the drag caused by incorporating resources. The parameter of importance here is p. In (3.10), $b - p$ can be greater than or less than 0. If the rate of natural resource utilization in the production process exceeds the rate at which government interventions improve the efficiency of natural resource utilization, then the growth rate of Y on the balanced growth path diminishes.

$$g_{Y^*} \begin{cases} < g_{\tilde{Y}} & (b - p) > 0 \\ = g_{\tilde{Y}} & (b - p) = 0 \\ > g_{\tilde{Y}} & (b - p) < 0 \end{cases} \tag{3.11}$$

where g_{Y^*} is (3.10) from e$g_{\tilde{Y}}$ is the BGP of basic Solow model. Equation (3.11) shows the influence of p (institutional quality) on the drag.

More on Drag

Intuitively so, the limitation of non-renewable resources seems to be the most apparent notion for drag. We can calculate the drag by subtracting the (3.10) from (3.6). Nordhaus, 1992 in his paper *Lethal model 2* examines concern about global economic growth limitation faced by limited resources and realized a .2 percentage point by which an economy is losing on growth each year due to these limitations. Another component is rent-seeking and qualitative factors like effective management of resources and investments in human resource development, which are difficult to incorporate. While the drag caused by the limitation of resources can be outweighed, partially so, by substituting for renewable resources through technological progress (Romer, 2012), the latter component needs more attention. Although realized, it is not clear how rent-seeking can be accounted for in our model with Solow assumptions. It has been executed by Torvik, 2002 (four-sector General Equilibrium model with Increasing returns to scale) and Wick et al., 2006 (two-stage contest and conflict model). Nonetheless, in view of these issues, empirically, one can tackle them by including institutional quality.

Shortcomings

One of the technical shortcomings of our GSM with the effectiveness of natural resources is that output plummets to zero if natural resources do not exist in the economy. Correspondingly, it does not inherently assume anything about the dependence debate. Further, it follows constant returns to scale (CRS), which is a simplification. This might have strong qualitative implications on natural resources as a variable, for it means that other inputs like technology and labour, need to offset the negative growth to be in CRS, which can be difficult.

Another aspect which needs attention is the difference between resource abundance and resource dependence. While we have discussed the influence of the institution on the resource-rich case, researchers

have found that while on one hand resource abundance has shown a significant impact on growth patterns, it is not the case in a resource-dependent economy Mehlum et al., 2006. Resource dependence of a country when the maximum share of natural resource rents in real GDP and resource abundance of a country is when the natural resource rent is small Ning Ding et al., 2005. In our model, we do not make a distinction between resource-abundant and resource-dependent. The model inherently does not project any specification but depends on the data. Although it is worth comparing the effects of natural resource abundance and dependence on growth parameters as the effect of these two concepts on economic growth is not necessarily the same (Brunnschweiler et al., 2008). The hypothesis discussed above of the negative trend of resource curse and positive influence of institutional quality is the result of the natural resource abundance case. Ning Ding et al., 2005, on the other hand, realized in their two model resource endowment and resource-dependent case, that while the former has a positive growth impact, the latter is significantly negative. They pointed out that an economy that is structurally dependent on the natural resource sector may have negative implications for growth. Further, it is observed that the measurement of resource dependence suffers from an endogeneity problem and thus should not be treated exogenously (Ploeg, 2011). Intuitively, a resource-dependent economy will imply rent-seeking activities, a lower level of democracy and an institutional rule of law that is unfavourable, thus seems unlikely to have a positive impact on growth.

4

Empirics

4.1 Threshold regression

In the attempt to analyze the threshold effect of institutional quality in resource curse discourse, the authors (Sarmidi et al., 2014) hypothesize that the influence of natural resources on economic growth becomes significant only once a certain threshold level of institutional quality is reached. Furthermore, the results suggest that countries with low institutional quality rely extensively on natural resources for growth, whereas nations with high-quality institutions exhibit less dependence on natural resources for generating economic growth. They use the threshold regression of the following form:

$$RGDPC_i = \begin{cases} \beta_0^1 + \beta_1^1 R_i + \beta_2^1 INS_i + \beta_3^1 X_i + e_i, INS \leq \lambda \\ \beta_0^1 + \beta_1^1 R_i + \beta_2^1 INS_i + \beta_3^1 X_i + e_i, INS > \lambda \end{cases} \tag{4.1}$$

where INS (i.e., level of institutional development) is the threshold variable used to split the sample into regimes or groups, and λ is the unknown threshold parameter. This type of modeling strategy allows the role of natural resources to differ depending on whether institutions are below or above some unknown level of λ.

The null hypothesis of linearity H0: $\beta_1 = \beta_2$ against the threshold model in equation (4.1). Further, it follows the heteroskedasticity-consistent Lagrange Multiplier (LM) bootstrap procedure to test the

null hypothesis of a linear formulation against a threshold regression alternative. To examine the possibility of high institutional quality as a driving force to a positive effect on growth in resource-rich countries, the equation is extended to include an interaction term between institutions and natural resources:

$$RGDPCi = \alpha_0 + \alpha_1 R_i + \alpha_2 INS + \alpha_3 (R \times INS)_i + \alpha_4 X_i + \epsilon_i \qquad (4.2)$$

The variable of importance is α_3. If α_3 is negative and statistically significant, the negative growth effect increases as institutional quality improves. On the other hand, if α_3 is positive and significant, this indicates that the negative growth effect diminishes as institutional quality improves. This study employs cross-country estimations to estimate equations with a sample dataset of 90 countries and the sample period spans from 1984 to 2005.

Table 4.1: *OLS and Threshold Regression Estimates of Equation* $RGDPCi = \alpha_0 + \alpha_1 R_i + \alpha_2 INS + \alpha_3 (R \times INS)_i + \alpha_4 X_i + \epsilon_i$

	Institutions: Government Effect		
	Linear	< 0.47	> 0.47
Constant	9.179***	52.15***	7.038**
	-2.05	-10.62	-3.26
$R^2 DPP_{1970}$	-1.568	-1.071	-1.671
	-0.383	-0.771	-0.331
Government effectiveness	7.875***	59.52***	7.869***
	(-1.541)	(-12.96)	(-1.891)
R-sq	0.4	0.389	0.392
No. Obs	77	21	56

(Source: Sarmidi et al., 2014)

Notes: Notes: The standard errors are reported in parentheses (White corrected for heteroskedasticity). Results correspond to a trimming percentage of 15%. *** and ** indicate significance at the 1% and 5% levels, respectively.

The results, from four models built with rule of law, government effect, corruption, and bureaucratic quality with threshold regression of λ show a negative coefficient for the natural resources, thence corroborating with the resource curse literature. Further, we can see that the interaction term between natural resources and institutional quality is positive and significant. In other words, institutional quality acts as an opposite force towards the curse trend. The curse can be weakened or even dodged, as the negative coefficient of natural resources can be reduced by the positive coefficient of the interaction term when the government effect gets stronger. If the government effect reaches a certain threshold level, then it will cancel out the effect of the resource curse. As discussed in the previous section, resource dependence suffers from the endogeneity problem as the data might have missing variables that can correlate with the resource-institution relation. Also, the robustness of the terms can be questioned and often leads to changed signs as a unique threshold level is required in different contexts.

Similar studies

Various studies have done a similar exercise by applying different threshold standards. Brunnschweiler, 2008 does a similar study by taking the export data, to challenge the resource curse literature. Their result shows that when the quality of institutions is controlled for, natural resources, and in particular mineral resources have a positive direct association with real GDP growth over the period 1970–2000. Sarmidi et al., 2014 uses the same data set but different threshold components to show that the institutional quality and growth correlation is non-linear and is positive only after a threshold. Further, Cabrales et al., 2011 estimates the threshold value by doing OLS regressions with fixed values and then choosing the value whose regression yields the minimum mean squared error following the Hansen framework. Their results show similar results where below the given threshold, the coefficient of interaction term between institutional quality is negative.

Lastly, Epo et al., 2020 works on African countries with panel data for threshold components of rule of law, control of corruption, and governance effectiveness. They check for results when a threshold is included and otherwise. When institutional quality is a threshold variable the relationship becomes positive compared to negative (curse) when not controlled for. Intuitively and in results, corruption shows a negative interaction coefficient.

4.2 Neglects of Education

Gylfason, 2001 in the attempt to theorize various channels of transmission to analyze the natural resource curse, focuses on the neglect of education and expenditure on human resources causing stunted economic development in resource-rich countries. Four main channels are the Dutch disease, rent-seeking, overconfidence, and neglect of education. The author finds out that the investments in education, measured by public expenditure, anticipated years of schooling, and gross secondary-school enrollment, display an inverse correlation with the proportion of natural capital within a nation's wealth across various countries. Further, as the share of natural resources in a country's wealth increases, the allocation towards human capital development through education decreases. As a consequence the progress of economic development is hindered and thus growth and development.

The author examines the relation with education through several regression analyses and observes various results. Figure 1 is the plot of annual growth on the share of natural capital in wealth, which shows a significant negative relationship parallel to the Natural resource curse literature. The data is time series data of 90 countries from 1980-97. Figure 2 plots public expenditure on education and natural resource abundance measured. Public expenditure on education acts as a yardstick at least to some extent for the government's commitment to education. According to the regression analysis of the 90 observations, an 18 percentage point rise in the share of natural capital from one

country to the next corresponds to a one percent decrease in public expenditure on education relative to Gross National Product (GNP). And the results are statistically significant. Qualitatively, this result provides a reason for poor countries locked in low-skill intensive natural-resource-based industries thus getting the curse. Figure 3 shows a plot of the expected number of years of schooling from 1980 to 1997 and natural resource abundance. The regression analysis conducted on the 91 observations indicates that for every five percentage point increase in the share of natural capital from one country to another, there is a corresponding decrease of ten percentage points in the secondary school enrollment rate. Decreasing years of schooling qualitatively could explain the poverty traps and other multi-dimensional poverty in countries. Figure 4 is a scatter plot of enrollment in secondary education and economic growth. The result shows an increase of 40 percentage points in secondary school enrollment is associated with a one percentage point rise in the annual rate of growth of Gross National Product (GNP) per capita.

In closing, while data shows economic growth varies inversely with natural resource abundance, the measures of education (expenditure on education and years of schooling) seem inversely related to natural resource abundance, and further, economic growth varies directly with education. Corresponding to this, Kronenberg, 2004 identifies the need to acknowledge the robustness of the regression results; as a negative correlation between natural resource abundance and economic growth can be attributed to different variables, in his case, to corruption. The coefficient tends to vary when different variables are included or excluded. Nonetheless, expenditures on education provide interesting insights into the impact on the growth and development in resource-abundant countries. The expenditures on human resource development, an essential domain of institutions and political economy, suggest that institutions can play a pivotal role in managing the resource curse.

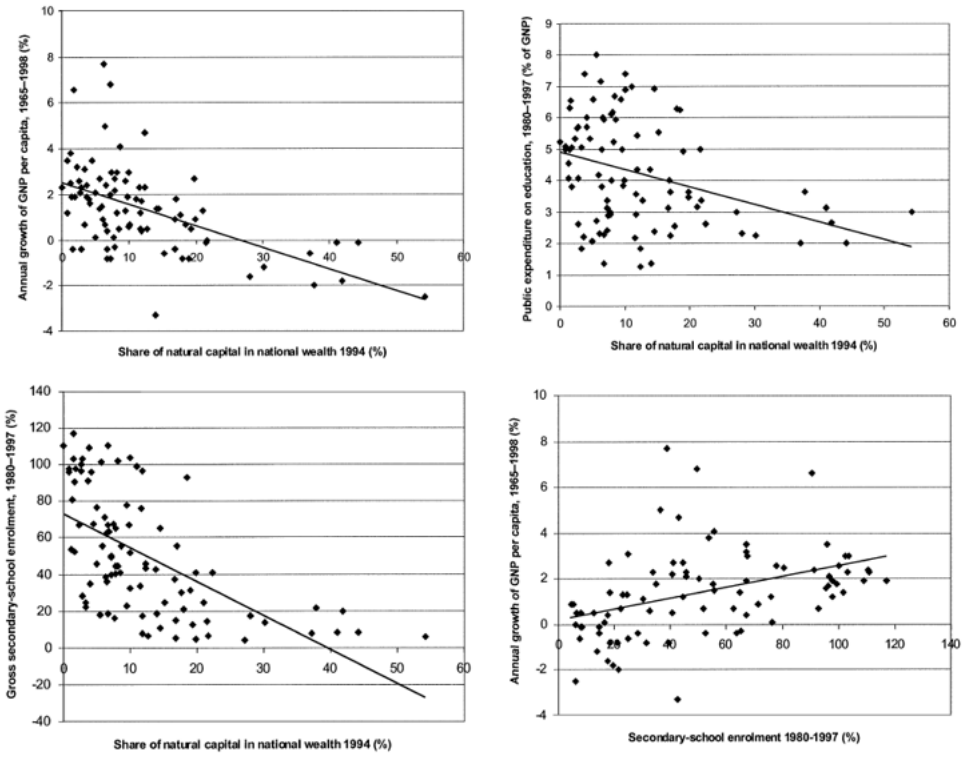

Figure 4.1: *Education and growth in resource abundance*

Gylfason, 2001

Fig.1 Economic growth and natural capital.(top left)
Fig.2 Expenditure on education and natural capital.(top right)
Fig.3 Secondary enrolment and natural capital.(left)
Fig.4 Economic growth and education. (right)

5

Conclusion

In this paper, we surveyed two major trends in resource curse literature, the negative growth trend and positive influence with the inclusion of institutional components. The focus, however, is the latter. We started with the basic Solow growth model and saw the effect of including natural resources (GSM). Further, we inserted another variable, the effectiveness of resources, as a proxy for institutional quality (similar to Romer, 2012, except they assume technology as the effectiveness factor). We further saw how the drag caused by resource limitation and unaccounted factors like rent-seeking can be reduced by including factors for effectiveness. In the empirical chapter, we studied how this hypothesis has been explored or rejected, through two major streams. First, the threshold regression analysis (Sarmidi et al., 2014) re-examines that the curse exists for a specific threshold level of institutional quality, after which it can turn into a blessing. The relation, although, is not linear as discovered. Second, the importance of expenditure on education as a factor in understanding the curse- realized by Gylfason, 2001. Emphasizing the risk pertaining to authorities neglecting education, the political institutions in resource-rich countries can, indeed, influence the curse hypothesis, for instance, through increasing expenditures on education.[1]

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